INCH-POUND
MIL-C-48482C(AR)
09 May 1995
(see 6.4)

# MILITARY SPECIFICATION

# CIRCUIT CARD ASSEMBLY - LASER CARD

# TEST LOAD SIMULATOR

This specification is approved for use by the U.S. Army Armament Research, Development and Engineering Center (ARDEC) and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Circuit Card Assembly Laser Card Test Load Simulator, 11732527 (see 6.1).

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, General Specification Governing the Manufacture and Inspection of

MIL-I-45607 - Inspection Equipment; Acquisition, Maintenance, and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use to improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-EDE-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

**AMSC N/A** FSC 1240

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STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-100 | - Engineering Drawing Practices |
| MIL-STD-454 | - Standard General Requirements for Electronic Equipment |
| MIL-STD-810 | - Environmental Test Methods |
| MIL-STD-2000 | - Standard Requirements for Soldered Electrical and Electronic Assemblies |
| MIL-STD-2073-1 | - Procedures for Development and Application of Packaging Requirements |

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issue shall be those in effect on date of the solicitation.

U.S. Army Armament Research Development and Engineering Center (ARDEC)

DRAWINGS

| | |
|---|---|
| 11732527 | - Circuit Card Assembly - Laser Card Test, Load Simulator |

(Copies of Government drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained (See contract provisions for additional precedence criteria).

3. REQUIREMENTS

3.1 Fabrication. The laser test load simulator circuit card assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing 11732527 and drawings pertaining thereto and, when assembled, shall meet the



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requirements of this specification (see 4.5.1).

3.1.1 Function. The assembly shall provide the following:

a. Deflection feedback (D Fdbk (IN)), elevation feedback (-E Fdbk (IN)) and laser deflection feedback (DL Fdbk (IN)) signals for simulation of the feedback inputs supplied to the M21 computer by the reticle projector, output and laser units.

b. Sense signals (CS1 (OUT) and CS2 (OUT)) for simulation of the outputs applied to the M21 computer by the cant unit.

c. Computer A10 card fail comparator logic for detecting a failure of the A10 card.

3.1.2 General specifications. (see 4.6.3)

3.1.2.1 Manufacture and inspection. The following requirements of MIL-F-13926 apply:

a. Dimensions and tolerances
b. Effect of protective coating on dimensions (Inorganic type coatings)
c. Part identification and marking
d. Workmanship (Including applicable portions of MIL-STD-2000 and MIL-STD-454, Requirement 9)

3.1.2.2 Standards of manufacture.

3.1.2.2.1 Assembly and soldering. The requirements of MIL-STD-2000 shall apply, as a minimum.

3.1.2.2.2 Interchangeability. The assembly shall be manufactured in accordance with interchangeability requirements as specified in MIL-STD-100.

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

| | | |
|---|---|---|
| a. | Temperature | 73° ± 18°F |
| b. | Relative humidity | 50 percent + 30 percent |
| c. | Atmospheric pressure | 28.5 + 2.0 -3.0 in. Hg. |

3.2 First article. When specified in the contract or purchase order (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.



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3.3.1 D Fdbk (IN) output. With the power sources (items 2.2 through 2.6 of Table I), the interconnections (item 4 of Table I), and the signal sources (items 3.1 through 3.4 of Table I) applied to the assembly; with the feedback circuit load (item 1.2 of Table I) connected between pins A26 and B5 and its return to pins A3, B3, as specified on Figure 1, and with pin A5 interconnected with pins A3, B3, and (see 4.6.1.1):

a. With switch S1 of Figure 1 closed, input signal (item 3.5 of Table I) at 4.700 ± 0.104V rms phase 1 applied to Vin of Figure 1, the output voltage at pin A26 measured with respect to pins A3, B3 shall increase from 0.0 ± 0.050V rms, 400 ± 1 Hz, phase 1 to 4.700 ± 0.500V rms, 400 ± 1 Hz, phase 1 and stabilize. With subsequent opening of switch S1, the output voltage at pin A26 shall decrease to 0.000 ± 0.050V rms within a period of 42 ± 16 seconds from the switch opening.

b. With switch S1 of Figure 1 closed, input signal (item 3.5 of Table I) at 4.700 ± 0.104V rms phase 2 applied to Vin of Figure 1, the output voltage at pin A26 measured with respect to pins A3, B3 shall increase from 0.000 ± 0.050V rms, 400 ± 1 Hz phase 2 to 4.700 ± 0.500V rms, 400 ± 1 Hz, phase 2 and stabilize. With subsequent opening of switch S1, the output voltage at pin A26 shall decrease to 0.000 ± 0.050V rms within a period of 42 ± 16 seconds from the switch opening.

3.3.2 DL Fdbk (IN) output. With the power sources (items 2.2 through 2.6 of Table I), the interconnections (item 4 of Table I), and the signal sources (items 3.1 through 3.4 of Table I) applied to the assembly; and with the feedback circuit load (item 1.2 of Table I) connected between pins A27 and A6 and to return pins A3, B3, as specified in Figure 1, with pin A8 connected to pins A3, B3, and (see 4.6.1.2):

a. With switch S1 of Figure 1 closed, input signal (item 3.5 of Table I) at 4.700 ± 0.104V rms phase 1 applied to Vin of Figure 1, the output voltage at pin A27 measured with respect to pins A3, B3 shall increase from 0.0 ± 0.050V rms, 400 ± 1 Hz, phase 1 to 4.700 ± 0.500V rms, 400 ± 1 Hz, phase 1 and stabilize. With subsequent opening of switch S1, the output voltage at pin A27 shall decrease to 0.000 ± 0.050V rms within a period of 5.1 to 11.4 seconds from the switch opening.

b. With switch S1 of Figure 1 closed, input signal (item 3.5 of Table I) at 4.700 ± 0.104V rms phase 2 applied to Vin of Figure 1, the output voltage at pin A27 measured with respect to pins A3, B3 shall increase from 0.000 ± 0.050V rms, 400 ± Hz phase 2 to 4.700 ± 0.500V rms, 400 ± 1 Hz, phase 2 and stabilize. With subsequent opening of switch S1, the output voltage at pin A27 shall decrease to 0.000 ± 0.050V rms within a period of 5.1 to 11.4 seconds from the switch opening.



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TABLE I. Loads, power, signals and interconnections.

| Item | Conditions | Characterisitics | Connections |
|---|---|---|---|
| 1.0 | Loads | | |
| 1.1 | Lamp | 24V, 50 ma lamp | When applied, connected between pins B6 (+) and A15 (-) |
| 1.2 | Feedback circuit (D Fdbk (IN) and DL Fdbk (IN)) | See Figure 1 | When aplied, connected between pins A26 and B5 or pins A27 and A6, as specified, and to return pins A3, B3 as shown on Figure 1. |
| 1.3 | Feedback circuit (E Fdbk (IN)) | See Figure 2 | When applied, connected between pins B26, B10, (B2 or A2), and return pins A3, B3 |
| 2.0 | Power sources | | |
| 2.1 | 24V dc | Tolerance: ±2V<br>Ripple 20mV p-p max<br>Current 500 ma min | Applied between pins B6(+) and (A23, B23) (-) |
| 2.2 | 15V dc | Tolerance: ±0.030V dc<br>Ripple 20mV p-p max<br>Current 500 ma min | Pins (A1, B1) (+) and (A3, B3) (-), and between pins as specified |
| 2.3 | 10V dc | Tolerance: ±0.010V dc<br>Ripple 20mV p-p max<br>Current 500 ma min | Pins (A7, B7) (+) and (A3, B3) (-) |
| 2.4 | 5.250V dc | Tolerance: ±0.150V dc<br>Ripple 20mV p-p max<br>Current 500 ma min | Pins (A4, B4) (+) and (A13, B13) (-) and between pin as specified |
| 2.5 | -10V dc | Tolerance: ±0.010V dc<br>Ripple 20mV p-p max<br>Current 500 ma min | Pins (A24, B24) (-) and (A3, B3) (+) |
| 2.6 | -15V dc | Tolerance: ±0.030V dc<br>Ripple 20mV p-p max<br>Current 500 ma min | Pins (A30, B30) (-) and (A3, B3) (+) and between pins as specified |



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TABLE I. Loads, power, signals and interconnections. (Continued)

| Item | Conditions | Characterisitics | Connections |
|---|---|---|---|
| 3.0 | Signal sources | | |
| 3.1 | 5V rms, 400 Hz | Tolerances: ±0.005V rms; ±1 Hz<br>Phase: 1 | When specified, applied to pins A22, B22, with return to pins A3, B3 |
| 3.2 | 4V rms, 400 Hz (AC REF 1) | Tolerances: ±0.005V rms; ±1 Hz<br>Phase: 2 (180° out-of phase with phase 1 in Item 3.1) | When specified, applied to pins A21, B21 with return to pins A3, B3 |
| 3.3 | 30 V p-p, 400 Hz (MR1) | Square wave<br>Tolerances: ±0.50V; ±1 Hz<br>Phase: 1 | Applied to pin B10 with return to pins A3, B3 |
| 3.4 | 30 V p-p, 400 Hz (MR2) | Square wave<br>Tolerances: ±0.50V; ±1 Hz<br>Phase: 2 (180° out-of phase with phase 1 in Item 3.3) | Applied to pin B8 with return to pins A3, B3 |
| 3.5 | Selectable: 0.0 to 11.0V rms 400 Hz | Tolerances: ±0.010V rms; ±1 Hz<br>Phase: 1 or 2 | Applied to pin A6 with return to pins A3, B3 |
| 3.6 | Selectable: 0.0 to 11.0V rms 400 Hz | Tolerances: ±0.010V rms; ±1 Hz<br>Phase: 1 or 2 | Applied to pin A8 with return to pins A3, B3 |
| 3.7 | Selectable: -10 to 10V dc | Tolerances: ±0.010V dc<br>Ripple 20mV p-p max<br>Current 500 ma min | Applied as specified with return to pins A3, B3 |
| 4.0 | Interconnections | | Interconnections of the following pins: |
| 4.1 | - | - | Pins A3, B3,<br>Pins A13, B13 and<br>Pins A23, B23 |



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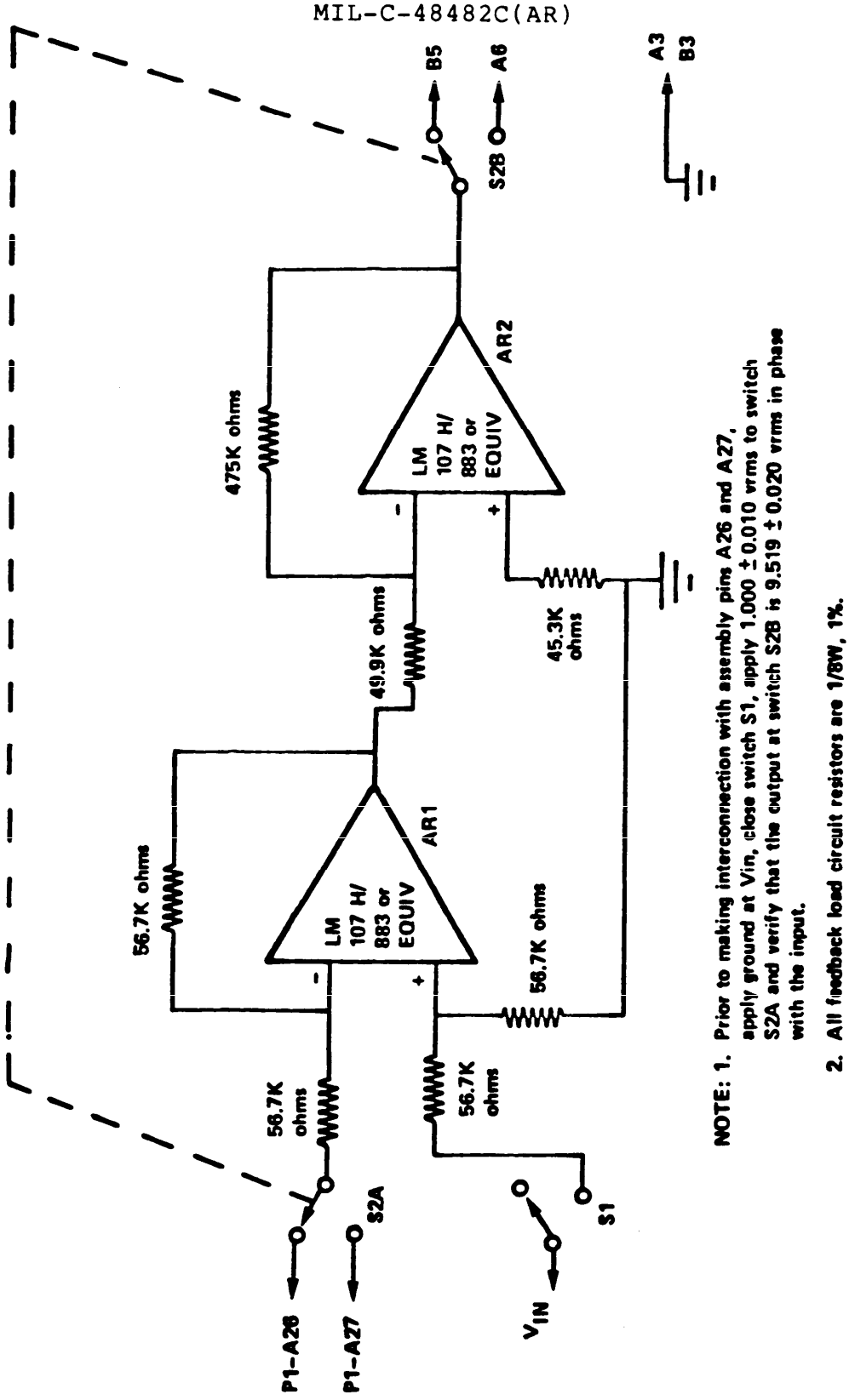



FIGURE 1. D Fdbk (IN) and DL Fdbk (IN) feedback load circuit and interconnection provisions.



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3.3.3 -E Fdbk (IN) output. With the power sources (items 2.2 through 2.6 of Table I), the interconnections (item 4 of Table I), and the signal sources (items 3.1 through 3.4 of Table I) applied to the assembly; and with the feedback circuit load (item 1.3 of Table I) connected to pins B26, B10, and A3, B3, as specified on Figure 2, and the output voltage at pin B26 measured with respect to pins A3, B3, the following requirements shall be met: (see 4.6.1.3)

3.3.3.1 Phase 1. With pin A2 grounded, connected to pins A3, B3, with pin B2 connected to terminal D, with switch S2 in position B, and:

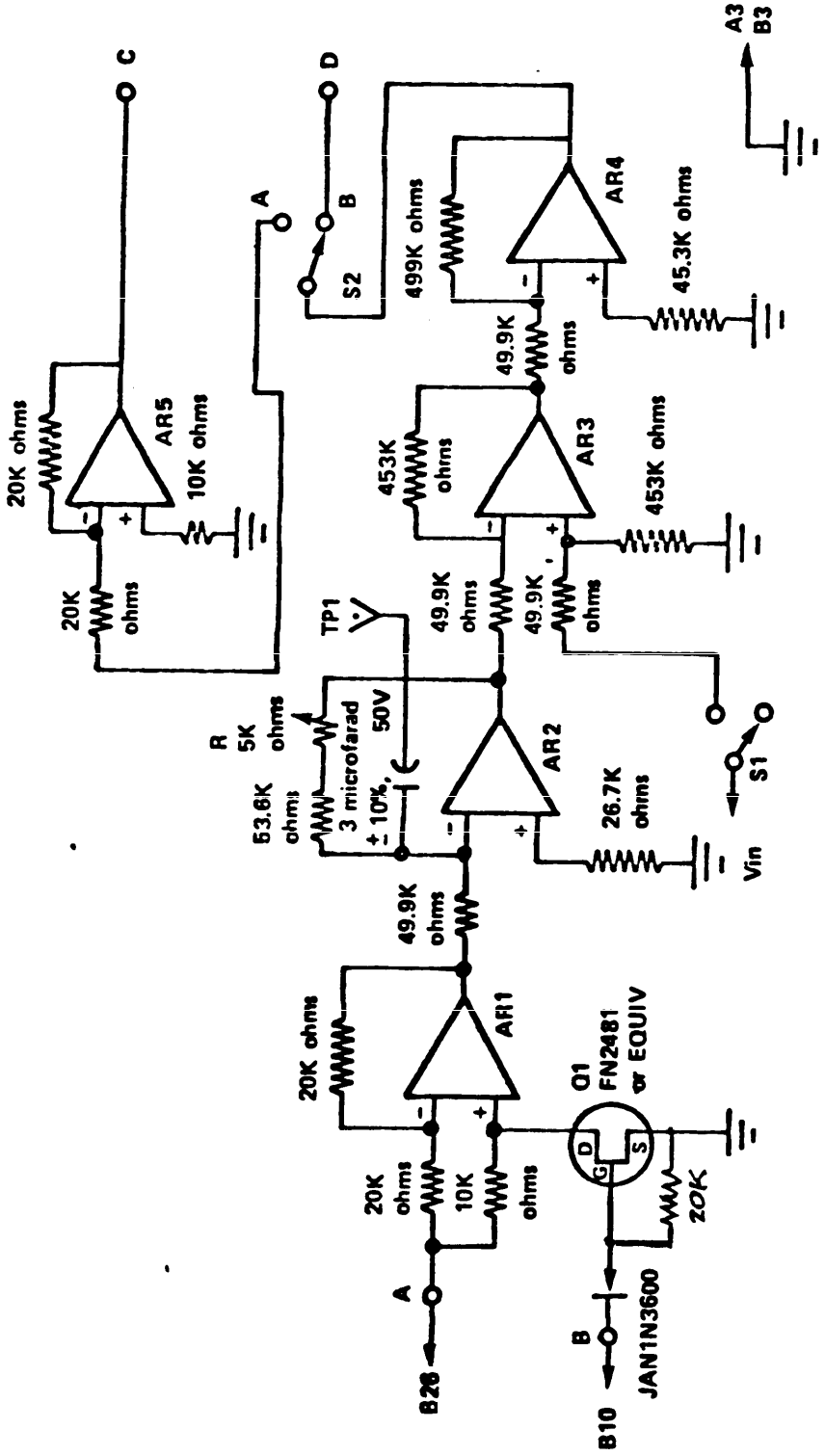
a. With switch S1 of Figure 2 closed and with the input signal (item 3.7 of Table I) at 2.000 ± 0.021V dc applied to Vin of Figure 2, the output voltage at pin B26 shall increase from 0.0 ± 0.050V rms, 400 ± 1 Hz phase 1, to 2.000 ± 0.500V rms, 400 ± 1 Hz phase 1, and stabilize. With subsequent opening of switch S1, the output voltage at pin B26 shall decrease to 0.000 ± 0.050V rms within a period of 40 ± 13 seconds from the switch opening.

b. With switch S1 of Figure 2 closed and with the input signal (item 3.7 of Table I) at -2.000 ± 0.021V dc applied to Vin of Figure 2, the output voltage at pin B26 shall increase from 0.0 ± 0.050V rms, 400 ± 1 Hz phase 2, to 2.000 ± 0.500V rms, 400 ± 1 Hz phase 2, and stabilize. With subsequent opening of switch S1, the output voltage at pin B26 shall decrease to 0.000 ± 0.050V rms within a period of 40 ± 13 seconds from the switch opening.

3.3.3.2 Phase 2. With pin B2 grounded (connected to pins A3, B3), with pin A2 connected to terminal C, with switch S2 in position A, with switch S1 of Figure 2 closed, and with the input signal (item 3.7 of Table I) at 2.000 ± 0.021V dc applied to Vin of Figure 2, the output voltage at pin B26 shall increase from 0.0 ± 0.050V rms, 400 ± 1 Hz phase 1, to 2.000 ± 0.500V rms, 400 ± 1 Hz phase 1, and stabilize. With subsequent opening of switch S1, the output voltage at pin B26 shall decrease to 0.000 ± 0.050V rms within a period of 40 ± 13 seconds from the switch opening.



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NOTE: 1. Prior to interconnecting with assembly pins B26 and B10, calibrate AR2 by applying ground to all ground terminals, applying 5.000 ± 0.010 vrms, 400 ± 1Hz, phase 1 to terminal A and applying 15.000 ± 0.500, 400 ± 1Hz, phase 1 square wave to terminal B, and adjusting R for ± 5.000 ± 0.010 vdc at TP1. Apply a ground at Vin, close switch S1, adjust the voltage at terminal A to 0.100 ± 0.010 vrms and verify that the voltage at the output of AR4 is 9.078 ± 0.020 vdc. Switch S2 to position A and verify AR5 output is -9.078 ± 0.020 vdc.

2. All feedback load circuit resistors are 1/8W, 1%.

3. AR1 through AR5 are LM107H/883 or equivalent.

FIGURE 2. -E Fdbk (IN) feedback load circuit, pre-connection calibration requirements, and interconnection provisions.



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3.3.4 LMTR1/LMTR2 comparator logic. With the power forms (items 2.1, 2.2, 2.4 and 2.6 of Table I), the 24V lamp load (item 1.1 of Table I) and signal sources (items 3.3 through 3.6 of Table I) applied to the assembly; and with pins A14, A16, B14 and B29 connected to logic return (Pins A13, B13) and with the 5.25V dc power source (item 2.4 of Table I) connected to pins A11 and B12, the following requirements shall be met (see 4.6.1.4).

3.3.4.1 Phase 1. With the input signal (item 3.6 of Table I) set to 0.000 ± 0.035V rms and the input signal (item 3.5 of Table I) adjusted as specified below, then:

a. With the input signal voltage adjusted to the nominal center point 0.000 ± 0.035V rms phase 1, the lamp shall extinguish.

b. With the input signal voltage increased from the nominal phase 1 voltage, the lamp shall illuminate. This change shall occur with the input signal voltage adjusted within the outward bound upper limit 5.778 ± 0.035V rms phase 1.

c. With the input signal voltage decreased toward zero volts, the lamp shall extinguish. This change shall occur with the input signal voltage adjusted within the inward bound upper limit 5.614 ± 0.348V rms phase 1.

d. With the input signal voltage decreased from the inward bound upper limit to a value of zero, the phase changed to phase 2, and the phase 2 voltage increased; the lamp shall illuminate. This change shall occur with the input signal voltage adjusted within the outer bound lower limit 6.090 ± 0.399V rms phase 2.

e. With the input signal voltage decreased toward zero volts, the lamp shall extinguish. This change shall occur with the input signal voltage adjusted within the inward bound lower limit 5.670 ± 0.348V rms phase 2.



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3.3.4.2 Phase 2. With the input signal (item 3.5 of Table I) set to 0.000 ± 0.035V rms and the input signal (item 3.6 of Table I) adjusted as specified below, then:

a. With the input signal voltage adjusted to the nominal center point 0.000 ± 0.035V rms phase 1, the lamp shall extinguish.

b. With the input signal voltage increased from the nominal phase 1 voltage, the lamp shall illuminate. This change shall occur with the input signal voltage adjusted within the outward bound upper limit 6.090 ± 0.399V rms phase 1.

c. With the input signal voltage decreased toward zero volts, the lamp shall extinguish. This change shall occur with the input signal voltage adjusted within the inward bound upper limit 5.670 ± 0.348V rms phase 1.

d. With the input signal voltage decreased from the inward bound upper limit to a value of zero, the phase changed to phase 2, and the phase 2 voltage increased, the lamp shall illuminate. This change shall occur with the input signal voltage adjusted within the outward bound lower limit 5.778 ± 0.352V rms phase 2.

e. With the input signal voltage decreased towards zero volts, the lamp shall extinguish. This change shall occur with the input signal voltage adjusted within the inward bound lower limit 5.614 ± 0.332V rms phase 2.

3.3.5 Computer A10 card fail logic. With the power forms (items 2.1, 2.2, 2.4 and 2.6 of Table I), the interconnections (item 4 of Table I), and the 24V lamp load (item 1.1 of Table I) applied to the assembly; and with the signal sources (items 3.3 and 3.4 of Table I) applied to the assembly, and item 3.5 applied and adjusted to 0.000 ± 0.035V rms and load lamp and the input conditions of Table II established in any item number order; the load lamp and output signal status shall be as specified in Table II (see 4.6.1.5).

3.3.6 CS2 (OUT) buffer amplifier. With the power sources (items 2.2 and 2.6 of Table I) and with the interconnections (item 4 of Table I) applied to the assembly; and with the signal input voltage (item 3.5 of Table I) applied at 2.500 ± 0.010V rms phase 1 to pins A20, B20; the output voltage at pin A25 measured with respect to pins A3, B3 shall be 2.500 ± 0.011V rms phase 2 (see 4.6.1.6).



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TABLE II. Computer A10 card fail logic.

| Item | Input at A8 1/ | Input Conditions at Pins 2/ | | | | | | Output Conditions 4/ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A11 | A14 | A16 | B12 | B14 | B29 | Lamp 3/ | A12 | B11 | B16 |
| 1 | 6.090 ± 0.399V rms, phase 1 | 1 | 1 | 1 | 1 | 0 | 0 | - | NA | 0 | 1 |
| 2 | 6.090 ± 0.399V rms, phase 1 | 0 | 1 | 1 | 1 | 0 | 0 | X | NA | 1 | 0 |
| 3 | 0.000 ± 0.010V rms | 1 | 1 | 1 | 1 | 0 | 0 | X | NA | NA | 0 |
| 4 | 0.000 ± 0.010V rms | 1 | 0 | 1 | 1 | 1 | 0 | X | NA | NA | 0 |
| 5 | 0.000 ± 0.010V rms | 1 | 0 | 0 | 1 | 0 | 0 | X | NA | NA | 1 |
| 6 | 6.090 ± 0.399V rms, phase 1 | 1 | 0 | 0 | 1 | 0 | 0 | X | NA | NA | 0 |
| 7 | 0.000 ± 0.050V rms | 0 | 0 | 0 | 1 | 0 | 0 | X | NA | NA | 0 |
| 8 | 0.000 ± 0.050V rms | 0 | 0 | 1 | 1 | 0 | 0 | - | NA | NA | 1 |
| 9 | NA | NA | 0 | 1 | 1 | 0 | 1 | X | 0 | NA | 0 |
| 10 | NA | NA | 0 | 1 | 0 | 0 | 1 | - | 1 | NA | 0 |

1/ Item 3.6 of Table I

2/ 1 = +5.25 ± 0.15V dc (Item 2.4 of Table I)
0 = 0.0 ± 0.5V dc interconnected with pin A13, B13

3/ X = illuminated. - = not illuminated.

4/ NA = indicates a not applicable condition

1 = +5.00 +0.50/-2.60
0 = 0.00 + 0.50



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3.3.7 CS1 (OUT) buffer amplifier. With the power sources (items 2.2 and 2.6 of Table I) and with the interconnections (item 4 of Table I) applied to the assembly; and with the signal input voltage (item 3.5 of Table I) applied to the assembly; and with the signal input voltage (item 3.5 of Table I) applied at 2.500 ± 0.010V rms phase 1 to pins A17, B17; the output voltage at pin A19 shall be 2.272 ± 0.011V rms phase 2 (see 4.6.1.7).

3.3.8 Phase 2, 3, and 5 switching. With the power forms (items 2.2 and 2.6 of Table I), the signal sources (items 3.1 and 3.2 of Table I), and the interconnections (item 4 of Table I) applied to the assembly; and with the Table III items input conditions established in any item number order, the corresponding ordered output conditions shall be as specified therein (see 4.6.1.8).

3.4 Environmental.

3.4.1 Temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of +40°F to 100°F and shall not be damaged when subjected to the storage conditions ranging from -65°F to 160°F (see 4.6.2.1 and 4.6.2.2).

3.4.2 Vibration. The assembly shall be capable of operating as specified herein after withstanding exposure under the conditions of 3.1.3 to a sinusoidal vibration displacement of .03 inch double amplitude, 10 to 55 Hz in accordance with Figure 3. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes (see 4.6.2.1).



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TABLE III. Phase 2, 3 and 5 switching.

| Item | Input Conditions At Pins 1/ (V dc) | | | | | | Output Voltage Forms At Pins 2/ | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | A18 | A28 | B18 | B28 | B19 | B25 | A26 | B26 | A27 | A19 |
| 1 | +15 | -15 | -15 | +15 | -15 | +15 | 1.180 ± 0.020 phase 2 | 0.682 ± 0.013 phase 1 | 2.261 ± 0.027 phase 1 | 1.988 ± 0.023 phase 1 |
| 2 | -15 | +15 | +15 | -15 | -15 | +15 | 0.682 ± 0.013 phase 2 | 1.180 ± 0.020 phase 2 | 1.997 ± 0.026 phase 1 | 1.217 ± 0.015 phase 1 |
| 3 | -15 | +15 | -15 | +15 | +15 | -15 | 0.227 ± 0.004 phase 1 | 1.508 ± 0.023 phase 2 | 0.217 ± 0.005 phase 2 | N/A 3/ |

1/ 15V dc = 15.000 ± 0.030 power form (Item 2.2 of Table I)
-15V dc = -15.000 ± 0.030 power form (Item 2.6 of Table I)

2/ Frequencies are 400 ± 1 Hz
phase 2 is 180° out-of-phase with phase 1

3/ N/A = not an applicable condition





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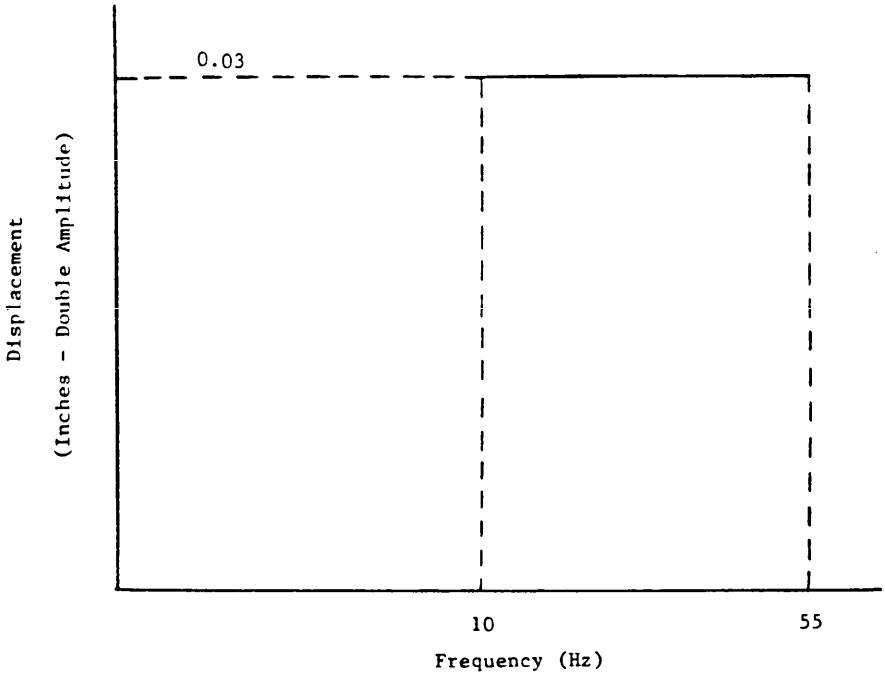



FIGURE 3. Vibration Curve.

3.4.3 Shock. The assembly shall be capable of operating as specified herein after withstanding exposure under the conditions of 3.1.3 to half sine wave shock pulses of 40 ± 4 gravity unit (g's) for a duration of 18 ± 3 milliseconds (msec) in accordance with Figure 4. The assembly shall withstand three shock pulses applied in both directions along three mutually perpendicular axes (18 shocks). In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 100.0 ± 10.0 g's for a duration of 1.5 ± 0.2 msec applied in each direction along three mutually perpendicular axes (see 4.6.2.1).

A = 40 g's
D = 18 msec

A = 100 g's
D = 1.5 msec

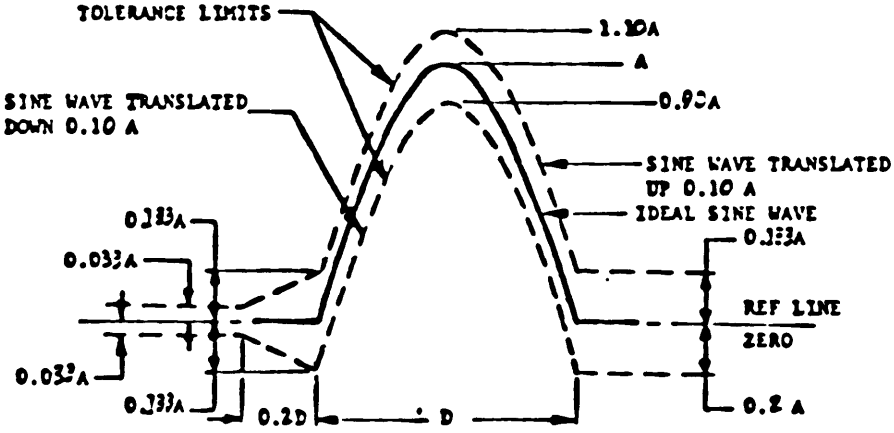



FIGURE 4. Shock Pulse.



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3.4.4 Temperature cycling. The assembly shall be capable of operating as specified herein after withstanding exposure to temperature cycling and soaking between the temperatures of -25°F and 140°F in accordance with Figure 5 (see 4.6.2.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in the specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and standard commercial practice. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract (see 6.5).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5 and 6.8).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article inspection. The requirements for first article approval and the designation of responsibility for first



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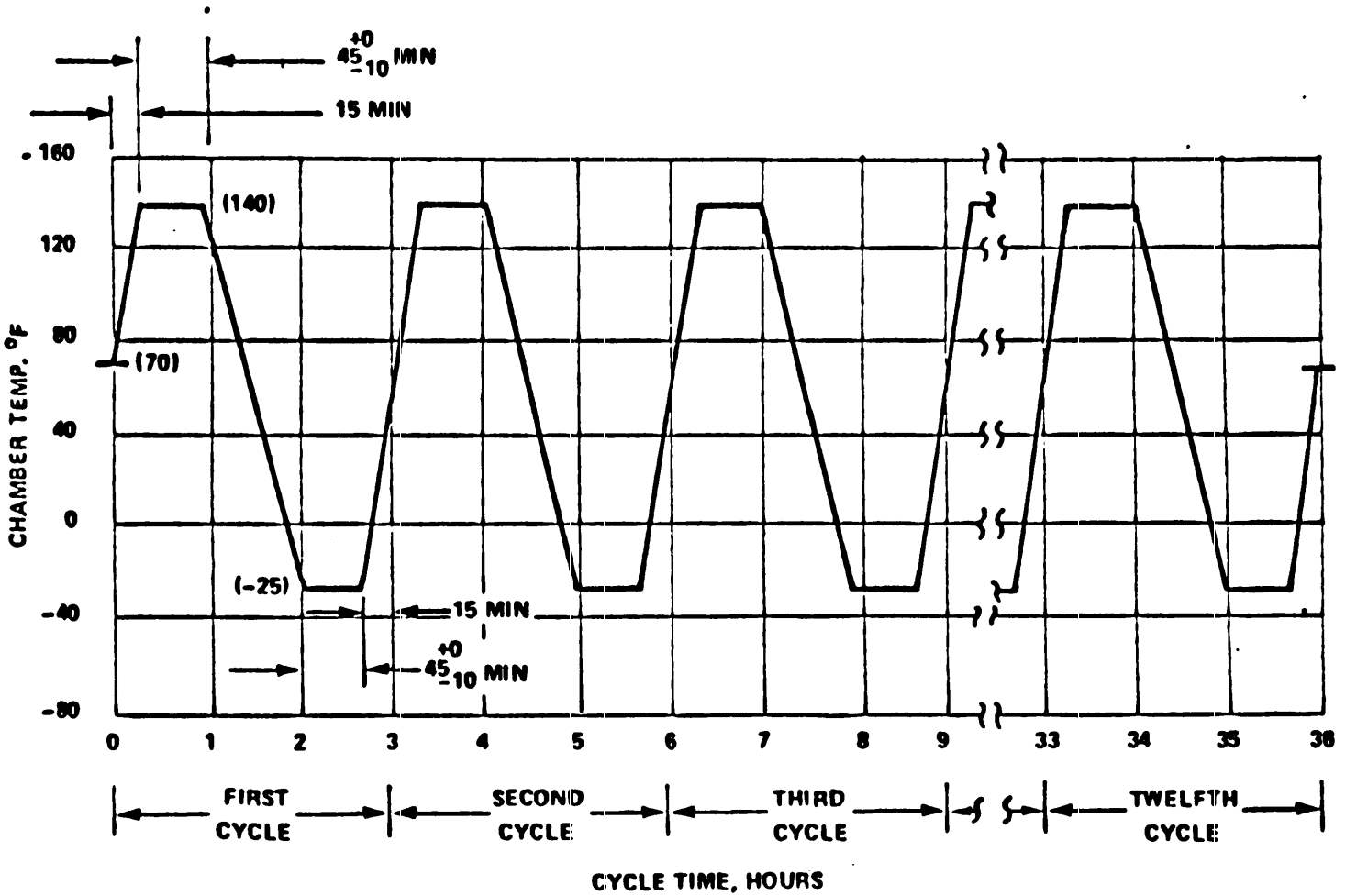



FIGURE 5. Temperature Cycling.



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article inspection to either the Government or the contractor shall be specified in the contract (see 6.3).

4.4.1 Sample. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.4.2. The first article sample shall consist of two assemblies.

4.4.2 Inspection. Two assemblies shall be subjected to all the inspections specified in Tables IV, V and VI. Tests 301 and 302 of Table VI shall be performed first.

TABLE IV. Performance tests.

| Item | Characteristics | Requirement | Test Procedure |
|---|---|---|---|
| 101 | D Fdbk (IN) output | 3.3.1 | 4.6.1.1 |
| 102 | DL Fdbk (IN) output | 3.3.2 | 4.6.1.2 |
| 103 | -E Fdbk (IN) output | 3.3.3 | 4.6.1.3 |
| 104 | LMTR1/LMTR2 comparator logic | 3.3.4 | 4.6.1.4 |
| 105 | Computer A10 card fail logic | 3.3.5 | 4.6.1.5 |
| 106 | CS2 (OUT) buffer amplifier | 3.3.6 | 4.6.1.6 |
| 107 | CS1 (OUT) buffer amplifier | 3.3.7 | 4.6.1.7 |
| 108 | Phase 2, 3 and 5 switching | 3.3.8 | 4.6.1.8 |

TABLE V. Environmental tests.

| Item | Characteristics | Requirement | Test Procedure |
|---|---|---|---|
| 201 | High temperature | 3.4.1 | 4.6.2.1 |
| 202 | Low temperature | 3.4.1 | 4.6.2.2 |
| 203 | Vibration | 3.4.2 | 4.6.2.3 |
| 204 | Shock | 3.4.3 | 4.6.2.4 |
| 205 | Temperature Cycling | 3.4.4 | 4.6.2.5 |

TABLE VI. General tests.

| Item | Characteristics | Requirement | Test Procedure |
|---|---|---|---|
| 301 | Fabrication | 3.1 | Applicable drawings - Visual |
| 302 | General specifications | 3.1.2 | 4.6.3 - Visual |
| 303 | Packaging inspection | 5.1 | 4.6.4 - Visual |



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4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any assembly to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

| Use | Plan |
|---|---|
| a. Logistics spare assembly apart from the next higher assembly | A, B and C |
| b. Component assembly of the next higher assembly | B and C |

4.5.2 Inspection plan A.

4.5.2.1 Environmental sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in Table V shall be applied separately to the assembly.

4.5.2.2 Acceptance. When the sample assembly of the quantity produced fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be resubmitted for reinspection only after all assemblies are re-examined or retested and all defective assemblies are removed or defects corrected.

4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative. All the tests in Table VI and only the test of item 205 in Table V shall apply. The following sampling plan (except that the visual examination for completeness,



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improper assembly and evidence of poor workmanship per items 301 and 302 shall be performed 100%) shall be utilized:

100%, c=0 for lot sizes of 10 or less
n=10, c=0 for lot sizes of 11 to 150
n=32, c=0 for lot sizes greater than 150

where, n=sample size and c=0 requires zero defects during inspection for acceptance

4.5.3.2 Acceptance. Where any one assembly in the sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be resubmitted as above.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in Table IV and shall be examined visually for completeness, improper assembly and evidence of poor workmanship per items 301 and 302 of Table VI. The conditions of 3.1.3 apply.

4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

4.6 Methods of inspection.

4.6.1 Performance tests.

4.6.1.1 D Fdbk (IN) output. Apply the power sources, interconnections and other input conditions of 3.3.1. Verify the output conditions as specified in 3.3.1.

4.6.1.2 $D_L$ Fdbk (IN) output. Apply the power sources, interconnections and other input conditions of 3.3.2. Verify the output conditions as specified in 3.3.2.

4.6.1.3 -E Fdbk (IN) output. Apply the power sources, interconnections and other input conditions of 3.3.3. Verify the output conditions as specified in 3.3.3.

4.6.1.4 LMTR1/LMTR2 comparator logic. Apply the power sources, interconnections and other input conditions of 3.3.4. Verify the output conditions as specified in 3.3.4.

4.6.1.5 Computer A10 card fail logic. Apply the power sources, interconnections and other input conditions of 3.3.5. Verify the logic output conditions as specified in 3.3.5.



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4.6.1.6 CS2 (OUT) buffer amplifier. Apply the power sources, interconnections and other input conditions of 3.3.6. Verify the logic output conditions as specified in 3.3.6.

4.6.1.7 CS1 (OUT) buffer amplifier. Apply the power sources, interconnections and other input conditions of 3.3.7. Verify the logic output conditions as specified in 3.3.7.

4.6.1.8 Phase 2, 3 and 5 switching. Apply the power sources, interconnections and other input conditions of 3.3.8. Verify the output conditions as specified in 3.3.8.

4.6.2 Environmental tests.

4.6.2.1 High temperature. Subject the assembly to the high temperature test specified in MIL-STD-810, Method 501 except storage temperature shall be 160°F for 48 hours. At the conclusion of the storage temperature test, stabilize the assembly at 100°F and subject it to the performance tests of Table IV. After testing, return the assembly to room temperature and subject it to the performance tests of Table IV.

4.6.2.2 Low temperature. Subject the assembly to the low temperature test specified in MIL-STD-810, Method 502. Maintain the temperature at -65°F for a period of 12 hours minimum. At the conclusion of the storage temperature test, stabilize the assembly at 40°F and subject it to the performance tests of Table IV. After testing, return the assembly to room temperature and subject the assembly to the performance tests of Table IV.

4.6.2.3 Vibration. Vibration A as specified in 4.6.2.3.1 shall be conducted only for first article inspection. Vibration B as specified in 4.6.2.3.2 shall be conducted only for quality conformance inspection.

4.6.2.3.1 Vibration A. Mount the assembly in a suitable vibration fixture and subject the assembly to the vibration test specified in MIL-STD-810, Method 514, General Procedures, and 3.4.2. The vibration amplitude shall have a tolerance of ±10%. The tolerance of the vibration frequency shall be ±2%, or ±1/2 Hz for frequencies below 25 Hz. Vibration sweeps shall be conducted with the duration of exposure not less than 80 minutes in each of three mutually perpendicular axes. At the conclusion of this test, subject the assembly to the performance tests of Table IV.

4.6.2.3.2 Vibration B. This test shall be conducted as in 4.6.2.3.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.2.4 Shock. Mount the assembly in a suitable shock fixture and subject it to the functional shock test for ground



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equipment specified in MIL-STD-810, Method 516, for a total of three half sine wave shock pulses in both directions along three mutually perpendicular axes (18 shocks). Peak amplitude shall be 40 ± 4 g's with a time duration of 18 ± 3 msec in accordance with Figure 4. At the conclusion of this test, subject the assembly to the performance tests of Table IV. Repeat this test with peak amplitude of 100.0 ± 10.0 g's with a time duration of 1.5 ± 0.2 msec in accordance with Figure 4.

4.6.2.5 Temperature cycling. With power and loads of Table I applied, subject the assembly to the temperature profile specified in Figure 5. At the conclusion of this test, subject the assembly to the performance tests of Table IV.

4.6.3 General specifications. Verification shall be performed by examination, visually or through the use of standard measuring equipment, in accordance with 3.1.2.

4.6.4 Packaging inspection. The sampling and inspection of the preservation, packaging, packing and container marking shall be in accordance with MIL-STD-2073-1.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with MIL-STD-2073-1. The level of protection shall be as specified in the procurement document (see 4.6.4 and 6.2d).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Laser Test Load Simulator circuit card (A6) assembly is a part of the Field Test Set, 11733200, which is used in the field for test of all of the M21 Computer subsystem components.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Inspection provisions for first article (see 6.3).

d. Applicable levels of preservation, packaging, packing, and container marking (see 5.1 and 6.7).



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6.3 First article. The contracting officer should provide specific guidance to offerors for the submission of a first article sample to be tested as specified in 4.4.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Supersession. Previous revisions of MIL-C-48482(AR) were for the US Army Armament, Research, Development and Engineering Center's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

6.5 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-QAC-F, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.6 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.7 Levels of packaging. The following definitions describe the three levels of packaging in accordance with MIL-STD-2073-1.

a. Level A - Adequate protection for indeterminate storage and worldwide distribution.

b. Level B - Adequate protection for known shipment conditions and storage conditions normally not exceeding one year.

c. Level C - Adequate protection for shipment to a known site for immediate use.



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6.8 Alternative quality conformance provisions. Alternative quality conformance procedures, methods, or equivalent, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equivalent, the contractor shall describe them in a written proposal submitted to the Government for evaluation. When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity.

6.9 Subject term (key word) listing.

Fire Control
Tank, M60A3
Test Set, Field, M21 Computer

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1240-0008)



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